

<page_start no = 1>

<sol_start id=1>

1. (3) 3

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<sol_start id=2>

2. (3) $\tan 60^\circ$

<sol_end>

<sol_start id=3>

3. (2) 30°

<sol_end>

<sol_start id=4>

4. (1) 10 cm

<sol_end>

<sol_start id=5>

5. (4) $a = 0$

<sol_end>

<sol_start id=6>

6. (4) Four decimal places

<sol_end>

<sol_start id=7>

7. (3) 3 and I

<sol_end>

<sol_start id=8>

8. (2) $(0, -1)$

<sol_end>

<sol_start id=9>

9. (4) No solution

<sol_end>

<sol_start id=10>

10. (1) $k = -9/2$

<sol_end>

<sol_start id=11>

11. (1) $k \neq -3$

<sol_end>

<sol_start id=12>

12. (1) 8

<sol_end>

<sol_start id=13>

13. (1) similar

<sol_end>

<sol_start id=14>

14. (1) $x^2y^4z^4$

<sol_end>

<sol_start id=15>

15. (3) $x^2/2 - x/2 - 6$

<sol_end>

<sol_start id=16>

16. (3) intersecting or coincident

<sol_end>

<sol_start id=17>

17. (1) Parallel

<sol_end>

<sol_start id=18>

18. (4) $16 : 81$

<sol_end>

<sol_start id=19>

19. (3) Assertion (A) is true but Reason (R) is false

<sol_end>

<sol_start id=20>

20. (1) Both A and R are true and R is correct explanation of A

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<sol_start id=21>

21. $6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5 \Rightarrow 6 \times 4 \times 3 \times 2 \times 1 \times 5(1 + 1)$
 $= 6 \times 4 \times 3 \times 2 \times 1 \times (2) \times 5$

Since it has more than two factors it is composite.

<sol_end>

<sol_start id=22>

22. $4x + 6y = 7 \Rightarrow 4x + 6y - 7 = 0$ ($a_1x + b_1y + c_1 = 0$)

$$kx + 18y = 21 \Rightarrow kx + 18y - 21 = 0 \text{ (} a_2x + b_2y + c_2 = 0 \text{)}$$

$$\text{for infinite solutions } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\text{So, } \frac{4}{k} = \frac{6}{18} = \frac{7}{21} \Rightarrow \frac{4}{k} = \frac{1}{3} \Rightarrow k = 12$$

$\therefore$ Value of k is 12.

<sol_end>

<sol_start id=23>

23. origin (0, 0) and point 2 is (-6, 8)

$$\text{By Distance formula} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$\Rightarrow \sqrt{(0 + 6)^2 + (0 - 8)^2}$$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100} = 10$$

So, distance is 10 units.

<sol_end>

<sol_start id=24>

24. Given : $\sin 3A = \cos (A - 10^\circ)$

we know that $\sin \theta = \cos (90 - \theta)$

So, $\sin 3A = \cos (90 - 3A)$

$$90 - 3A = A - 10 \text{ (from given)}$$

$$100 = 4A \Rightarrow A = 25^\circ \text{ and } 3A = 75^\circ$$

So, $3A$ is an acute angle.

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<sol_start id=25>

25. Given : $AB \parallel CD$

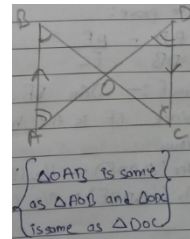
$$\angle B = \angle C \text{ (Alternate Angle)}$$

$$\angle A = \angle D \text{ (Alternate Angle)}$$

Proved above information for $\triangle OAB$ and $\triangle ODC$

By AA rule

$$\therefore \triangle AOB \sim \triangle DOC$$



<sol_end>

<sol_start id=26>

$$26. \quad \frac{\cos(90-\theta)}{1+\sin(90-\theta)} + \frac{1+\sin\theta(90-\theta)}{\cos(90-\theta)} = 2\operatorname{cosec}\theta$$

$$= \frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} \quad \left\{ \begin{array}{l} \sin(90-\theta) = \cos\theta \\ \cos(90-\theta) = \sin\theta \end{array} \right.$$

$$= \frac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta(1+\cos\theta)}$$

$$= \frac{1-\cos^2\theta + 1+\cos^2\theta + 2\cos\theta}{\sin\theta(1+\cos\theta)} \quad \left\{ \begin{array}{l} \sin^2\theta = 1-\cos^2\theta \\ (\sin^2\theta + \cos^2\theta = 1) \end{array} \right.$$

$$= \frac{2+2\cos\theta}{\sin\theta(1+\cos\theta)} \Rightarrow \frac{2(1+\cos\theta)}{\sin\theta(1+\cos\theta)} = \frac{2}{\sin\theta}$$

$$= 2 \operatorname{cosec} \theta \quad \{1/\sin \theta = \operatorname{cosec} \theta\}$$

Hence proved

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<sol_start id=27>

27. Base proportionality theorem states that, if a line is parallel to a side of a triangle which intersects the other sides into two distinct points, then the line divides those sides in proportion for that triangle.

Let $\triangle ABC$, where line l is parallel to BC , intersect AB at D and AC at E

To prove :

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Proof : - Join BE , CD and Draw $EF \perp AB$, $DG \perp CA$

Since EF is the height of $\triangle ADE$ and $\triangle DBE$

Now,

$$\text{Ar}(\triangle ADE) = \frac{1}{2} (\text{Base} \times \text{Height}) = \frac{1}{2} \times AD \times EF$$

$$\text{Ar}(\triangle DBE) = \frac{1}{2} (\text{Base} \times \text{Height}) = \frac{1}{2} \times DB \times EF$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DBE)} = \frac{1/2 \times AD \times EF}{1/2 \times DB \times EF} = \frac{AD}{DB} \quad \dots\dots(i)$$

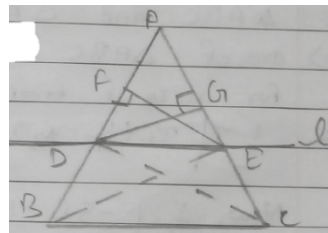
Also,

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DCE)} = \frac{1/2 \times AE \times DG}{1/2 \times EC \times DG} = \frac{AE}{EC} \quad \dots\dots(ii)$$

But $\triangle DBE$ and $\triangle DCE$ are on the same base DE and between the same parallel

$$\therefore \text{ar}(\triangle DBE) = \text{ar}(\triangle DCE) \quad \dots\dots(iii)$$

From (i), (ii) and (iii) we have



$$\frac{AD}{DB} = \frac{AE}{EC}$$

Hence proved

<sol_end>

<sol_start id=28>

28. let Sagar age be x and Tiru age be y

Given

$$x - 5 = 2(y - 5) \quad \dots\dots(i)$$

$$x + 10 - (y + 10) = 10 \quad \dots\dots(ii)$$

$$\text{So, } x - 5 = 2y - 10 \Rightarrow x - 2y = -5$$

$$x + 10 - y - 10 = 10 \Rightarrow x - y = 10$$

New, By elimination

$$x - 2y = -5$$

$$- \underline{x - y = 10}$$

$$-y = -15 \Rightarrow y = 15$$

$$\text{Now, } x - 2(15) = -5 = x - 30 - 5 \Rightarrow x = 25$$

So, age of Sagar is 25 year

Age of Tiru is 15 years

When Tiru have borned Saga's age was 10 years.

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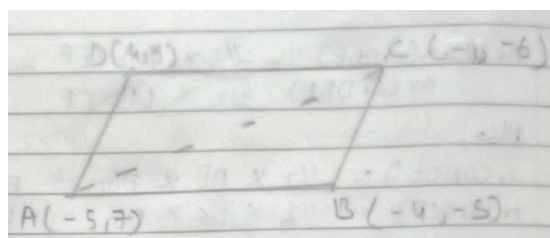
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29. Given

$$A(-5, 7)$$

$$B(-4, -5)$$



C (-1, -6)

D (4, 5)

To find its area we will split it into 2 triangles ΔABC and ΔACD

$\Rightarrow$ area of ΔABC

For area of triangle we have :-

$$= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$\text{Now, } \frac{1}{2} |-5(-5 - (-6)) + (-4)(-6 - (7)) + (-1)(7 - (5))|$$

$$= \frac{1}{2} |-5(1) + (-4)(-13) + (-1)(12)|$$

$$= \frac{1}{2} |-5 + 52 - 12| = \frac{1}{2} |35| = 17.5 \text{ units square}$$

Now for ΔACD

$$= \frac{1}{2} |-5(-6 - (5)) + (-1)(5 - (7)) + 4(7 - (-6))|$$

$$= \frac{1}{2} |-5(-11) + (-1)(-2) + 4(13)|$$

$$= \frac{1}{2} |55 + 2 + 52|$$

$$= \frac{1}{2} |109| = 54.5 \text{ units square}$$

So, Total area is 72 sq units.

<sol_end>

<sol_start id=30>

30. MCF of 861 and 1353 -

$$1353 = 861 \times 1 + 492$$

$$861 = 492 \times 1 + 369$$

$$392 = 369 \times 1 + 123$$

$$369 = 123 \times 3 + 0$$

So, HCF of 861 and 1353 is 123.

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<sol_start id=31>

31. A point of x-axis means y-co-ordinate is 0

So, it is in form of (x, 0)

Given

Distance of (x, 0) from (5, -2) and (-3, 2) is equal

By distance formulas

$$\Rightarrow \sqrt{(5-x)^2 + (-2-0)^2} = \sqrt{(-3-x)^2 + (2-0)^2}$$

$$\Rightarrow \sqrt{(5-x)^2 + 4} = \sqrt{(-3-x)^2 + 4}$$

$$\Rightarrow (5-x)^2 + 4 = (-3-x)^2 + 4$$

$$\Rightarrow (5-x)^2 = (-3-x)^2$$

$$\Rightarrow 25 + x^2 - 10x = 9 + x^2 + 6x$$

$$16 = 16x \Rightarrow x = 1$$

So, value of x is 1

<sol_end>

<sol_start id=32>

32. $\tan A + \sin A = m$

$\tan A - \sin A = n$

$$m^2 = (\tan A + \sin A)^2 = \tan^2 A + \sin^2 A + 2\sin A \tan A$$

$$n^2 = (\tan A + \sin A)^2 = (\tan^2 A + \sin A - 2 \sin A \tan A$$

$$m^2 - n^2 = (\tan A + \sin A)^2 - (\tan A - \sin A)^2$$

$$= 4 \sin A \tan A \quad \dots\dots(i)$$

$$\text{Now, } mn = (\tan A + \sin A) (\tan A - \sin A)$$

$$= \tan^2 A - \sin^2 A$$

$$= \sin^2 A / \cos^2 A - \sin^2 A = \frac{\sin^2 A - \sin^2 \cos^2 A}{\cos^2 A}$$

$$\frac{\sin^2 A (1 - \cos^2 A)}{\cos^2 A} \Rightarrow \frac{\sin^2 A (\sin^2 A)}{\cos^2 A} \quad \{1 - \cos^2 \theta = \sin^2 \theta\}$$

$$\frac{\sin^4 A}{\cos^2 A} = mn \text{ and root } mn = \frac{\sin^2 A}{\cos A} \quad \dots\dots(ii)$$

From (i)

$$\sin A \cdot \tan A = \sin A \times \sin A / \cos A = \sin^2 A / \cos A \quad \dots(ii)$$

So, LHS

$$= 4 \sin A \cdot \tan A \Rightarrow 4\sqrt{mn} \quad \text{from (iii)}$$

Hence proved

<sol_end>

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<sol_start id=33>

33. Given

$$x - y + 1 = 0 \quad \dots\dots(i)$$

$$3x + 2y - 12 = 0 \quad \dots\dots(ii)$$

Now,

$$x - y = -1 \text{ and } 3x + 2y = 12$$

for the (i) equation.

$$x = -1 + y \Rightarrow x = y - 1$$

x	-1	0	1	2	3
y	0	1	2	3	4

For the (ii) equation

$$3x + 2y = 12 \Rightarrow 3x = 12 - 2y \Rightarrow x = 4 - 2/3y$$

x	4	10/3	8/3	2
Y	0	1	2	3

x, y intersects at 2, 3

Points on x - axis are : (-1, 0) and (4, 0)

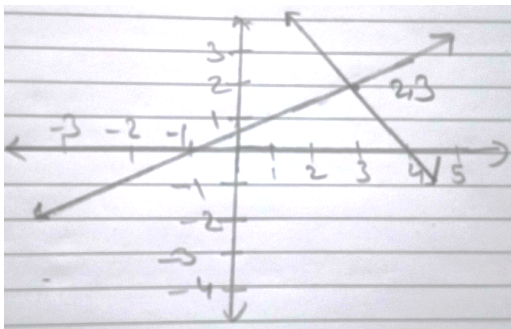
So the area of triangle formed by it is :-

$$\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$\frac{1}{2} |-1(3-0) + 2(0-0) + 4(0-3)|$$

$$\frac{1}{2} |(-3) + 0(-12)| = \frac{1}{2} |-15| = \frac{1}{2} (15)$$

Area of triangle formed by intersection and x-axis is 7.5 sq units.



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34. Given :

$$\text{Polynomial} = (2x^3 - 4x - x^2 + 2)$$

Two zeros are $\sqrt{2}$ and $-\sqrt{2}$

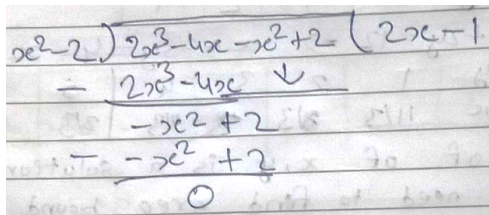
So,

we and write

$x - \sqrt{2} = 0$ and $x + \sqrt{2} = 0$ as they are zero/roots

Now, $(x - \sqrt{2})(x + \sqrt{2}) = x^2 - 2$

Now,


$$\begin{array}{r} x^2 - 2 \overline{) 2x^3 - 4x - x^2 + 2} \\ \underline{2x^3 - 4x} \\ -x^2 + 2 \\ \underline{-x^2 + 2} \\ 0 \end{array}$$

So, $(2x - 1)(x - \sqrt{2})(x + \sqrt{2}) = 2x^3 - 4x - x^2 + 2$

Now, $2x - 1 = 0 \Rightarrow 1 = 2x \Rightarrow \frac{1}{2} = x$

Therefore its all zeros are

$$\frac{1}{2}, \sqrt{2}, -\sqrt{2}$$

Since it has degree of 3 it have 3 zeros.

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<sol_start id=35>

35. Given :

Point A = (4, -3)

Point B = (8, 5)

Ratio is which they divide 3 : 1

We need to find Point C (Point, which divides the line segment in 3 : 1)

So,

By Section formula

$$\left[\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} \right), \left(\frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right) \right] = C(x, y)$$

$$m_1 = 3, m_2 = 1, x_1 = 4, x_2 = 8, y_1 = -3, y_2 = 5$$

Now,

$$\left[\left(\frac{3(8) + (1)(4)}{3+1} \right), \left(\frac{3(5) + 1(-3)}{3+1} \right) \right] = \text{Point C}(x, y)$$

$$\left[\left(\frac{24+4}{4} \right), \left(\frac{15+3}{4} \right) \right] = C(x, y)$$

$$= [7, 3] = C(x, y)$$

So the co-ordinates for C are (7, 3)

<sol_end>

<sol_start id=36>

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36.(i) (2) (0, 12)

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<sub_id = 2>

(ii) (1) (3, 5)

<sub_id end>

<sub_id = 3>

(iii) (3) 5m

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<sol_end>

<sol_start id=37>

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37. (i) (3) Parabola

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<sub_id = 2>

(ii) (3) 0

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<sub_id = 3>

(iii) (3) -3, 2

<sub_id end>

<sol_end>

<sol_start id=38>

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38. (i) (3) 24 cm^2

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(ii) (1) RHS

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<sub_id = 3>

(iii) (4) Converse of Pythagoras Theorem

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